SOFTWARE TEST CASES & RESULTS

Test ID	Requirement ID	Feature	Precondition	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Date	Evidence
TC-001	FR-01	Student Login	Student account exists and system is online	1. Open login page. 2. Enter valid student credentials. 3. Click Login.	Valid student username and password	Student should successfully log in and be redirected to dashboard.	Student logged in successfully and dashboard displayed.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-002	FR-10	Student Dashboard	Student is logged in	1. Log in as student. 2. View dashboard.	Student account	Dashboard should load and all functions should be accessible.	Dashboard appeared and worked correctly.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-003	FR-09	View Existing Logbook Entry	Student has an existing logbook entry	1. Open dashboard. 2. Click on an existing entry date.	Existing logbook entry	Selected entry details should open successfully.	Entry displayed successfully.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-004	FR-09, FR-10	Back to Logbook Navigation	Student is viewing a logbook entry	1. Open an entry. 2. Click "Back to Logbook".	Existing entry	User should be redirected to Logbook Dashboard.	Redirected successfully to Logbook Dashboard.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen
TC-005	FR-04	Create New Logbook Entry	Student is logged in	1. Click "New Entry". 2. Complete all fields. 3. Submit entry.	Logbook information (date, activities, hours worked, comments)	New entry form should open and accept user input.	New entry created successfully and all fields were functional.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-006	FR-04, FR-10	Save Logbook Entry	New logbook entry has been completed	1. Fill in entry form. 2. Click "Save Entry".	Completed logbook details	Entry should save successfully and appear on dashboard.	Entry saved successfully and displayed on logbook dashboard.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-007	FR-14	Student Sign Out	Student is logged in	1. Click "Sign Out".	N/A	User should be logged out and redirected to login page.	Redirected to login page successfully.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-008	FR-01	Coordinator Login	Coordinator account exists	1. Open login page. 2. Enter valid coordinator credentials. 3. Click Login.	Valid coordinator username and password	Coordinator should successfully log in.	Coordinator signed in successfully.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-009	FR-11	Coordinator Dashboard	Coordinator is logged in	1. Log in as coordinator. 2. View dashboard.	Uploaded student logbook entries	Dashboard should display students who submitted entries.	Submitted student entries were visible.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-010	FR-14	Coordinator Sign Out	Coordinator is logged in	1. Click "Sign Out".	N/A	User should be redirected to login page.	Redirected successfully to login page.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-011	FR-01	Student Login with Invalid Credentials	Student login page is accessible	1. Open login page. 2. Enter invalid username and/or password. 3. Click Login.	Invalid username/password	System should display an error message and deny access.	Error message displayed and login was unsuccessful.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-012	FR-04	Save Logbook Entry with Required Fields Empty	Student is logged in and on the New Entry page	1. Click "New Entry". 2. Leave one or more required fields blank. 3. Click "Save Entry".	Incomplete logbook entry	System should validate input and prevent saving until all required fields are completed.	Validation message displayed and entry was not saved.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-013	FR-01	Coordinator Login with Invalid Credentials	Coordinator login page is accessible	1. Enter incorrect login credentials. 2. Click Login.	Invalid coordinator username/password	System should display an error message and deny access.	Error message displayed and login was unsuccessful.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-014	FR-04, NFR-04	Verify Saved Data After Logout and Login	Student has previously saved a logbook entry	1. Create and save a logbook entry. 2. Sign out. 3. Sign back in. 4. Open the logbook dashboard.	Previously saved logbook entry	Saved entry should still be available on the dashboard.	Saved entry remained visible after logging back in.	Pass	Sihle Kayise	2026/10/22	Manual test in Chrome (screen check)
TC-110	FR-01	Sign in (student)	Seeded student account exists	1. Send the sign-in form with the email in capital letters and the correct password.	STUDENT2@ctucareer.co.za / student123	Signed in; session cookie set; redirected to /student.	redirect=/student; session cookie set	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-111	FR-01	Sign in with wrong password	Seeded student account exists	1. Send the sign-in form with a wrong password.	student2@ctucareer.co.za / wrong-password	Message 'Invalid email or password.'; no session cookie.	message="Invalid email or password."; cookie not set	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-112	FR-01, FR-03	Sign in (coordinator and lecturer)	Seeded staff accounts exist	1. Sign in as coordinator. 2. Sign in as lecturer.	coordinator@ / coord123; lecturer@ / lecturer123	Both are redirected to /coordinator.	coordinator -> /coordinator; lecturer -> /coordinator	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-113	FR-02	Sign-up (valid)	Email not yet registered	1. Send the sign-up form with a valid CTU student email and a strong password.	<year>0199@ctucareer.co.za / Strong#Pass1	Student account created; signed in; redirected to /student.	redirect=/student; user row role=student	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt

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TC-114	FR-02	Sign-up password rules	Sign-up page open	1. Try five passwords that each break one rule.	short1!, lowercase1!, UPPERCASE1!, NoNumbers!!, NoSpecial12	Each attempt is rejected with the message for the broken rule.	short1! -> "Password must be at least 8 characters."; lowercase1! -> "Password must include at least one uppercase letter."; UPPERCASE1! -> "Password must include at least one lowercase letter."; NoNumbers!! -> "Password must include at least one number."; NoSpecial12 -> "Password must include at least one special character."	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-115	FR-02	Sign-up email rules	Sign-up page open	1. Try a Gmail address. 2. Try a malformed CTU address.	someone@gmail.com; 1999@ctucareer.co.za	Both rejected with 'Please use a valid CTU email.'	gmail -> "Please use a valid CTU email."; 1999@ctucareer.co.za -> "Please use a valid CTU email."	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-116	FR-02	Sign-up duplicate email	TC-113 account exists	1. Sign up again with the same email in capital letters.	Same email, upper case	Rejected with 'An account with this email already exists.'	message="An account with this email already exists."	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-117	FR-03	Signed-out access	Not signed in	1. Open /student. 2. Open /coordinator.	No cookie	Both redirect (307) to the sign-in page.	/student -> 307 /; /coordinator -> 307 /	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-118	FR-03	Coordinator blocked from student area	Signed in as coordinator	1. Open /student.	Coordinator session	Redirected to /coordinator; logbook not shown.	status 307; redirected to /coordinator: true	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-119	FR-03	Student blocked from dashboard	Signed in as student	1. Open /coordinator.	Student session	Redirected to /student; dashboard not shown.	status 307; redirected to /student: true	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-120	FR-03	Student cannot view another student's entry	Signed in as Sihle	1. Open /student/entries/entry-1 (Dylan's entry).	entry-1	404 Not Found; entry text not shown.	status 404; entry text shown: false	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-120b	FR-03	Student cannot submit another student's entry	Signed in as Sihle	1. Call Submit for verification on entry-1.	entry-1	Rejected with 'Entry not found.'	message="Entry not found."	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-121	FR-03, NFR-03	Forged / tampered session cookie (DEF-01 retest)	DEF-01 fix deployed	1. Open /coordinator with cookie 'coordinator-1'. 2. Open it with 'coordinator-1' plus another user's signature.	Forged cookies	Both rejected; dashboard not shown.	unsigned 'coordinator-1' -> dashboard shown: false; another user's signature -> dashboard shown: false	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-122	FR-04	Create entry	Signed in as student	1. Submit the new-entry form with valid values.	2026-10-22, description, 4.5 h, mentor email	Entry saved with status Draft and opened.	redirect=/student/entries/631b4788-c53e-4eb8-8da2-aeca1b8c00b5; status=draft; hours=4.5	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-123	FR-04	Entry validation	Signed in as student	1. Try hours 30, 0 and -2. 2. Try an empty description.	Invalid hours / empty description	Each attempt rejected with a message; nothing saved.	hours=30 desc=yes -> "Hours for one day cannot be more than 24."; hours=0 desc=yes -> "Work date, description, and positive hours are required."; hours=-2 desc=yes -> "Work date, description, and positive hours are required."; hours=5 desc=empty -> "Work date, description, and positive hours are required."	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-124	FR-05	Evidence upload (PNG)	Signed in as student	1. Create an entry with a PNG file attached.	photo.png	Entry saved; file stored under a random UUID name.	stored as e2f009fa-ad72-418b-99a5-ffad6f2bf8c9.png	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-125	FR-05	Evidence upload (blocked type)	Signed in as student	1. Create an entry with an .exe file attached.	virus.exe	Rejected with 'Evidence must be a PDF or image file.'	message="Evidence must be a PDF or image file."	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-126	FR-13	Evidence download access	TC-124 file exists	1. Download as owner, another student, signed-out visitor and coordinator. 2. Try a path-traversal file name.	Stored evidence file	Owner 200, other student 403, signed out 401, coordinator 200, traversal 404.	owner 200; other student 403; signed out 401; coordinator 200; path traversal attempt 404	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-127	FR-06	Submit for verification	Draft entry with mentor email	1. Submit the entry for verification.	TC-122 entry	48-character link valid for 7 days; status Pending verification.	token length=48; status=pending_verification; expires after 7.00 days	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-128	FR-06	Submit without mentor email	Draft entry without mentor email	1. Submit the entry for verification.	Entry with empty mentor email	Rejected with 'Add a mentor email ...'.	message="Add a mentor email on the entry before submitting."	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-129	FR-06	Resubmit with active link	Entry pending with a valid link	1. Submit the same entry again.	TC-127 entry	Rejected; still only one link for the entry.	message="Only draft or rejected entries, or entries whose link has expired, can be submitted."; tokens for entry=1	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt

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TC-130	FR-07	Mentor page without sign-in	Valid link from TC-127	1. Open /verify/<token> without a session.	Token from TC-127	Page shows student, date, hours, description and Approve / Reject.	status 200; shows student name, description and Approve/Reject buttons	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-131	FR-07, FR-12	Mentor approves	Valid link	1. Click Approve.	Token from TC-127	Entry Verified; verification time stored.	status=verified; verified_at=2026-10-22T22:54:02.000Z	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-132	FR-07, FR-10	Mentor rejects; student resubmits	New entry submitted	1. Mentor rejects. 2. Student submits again.	New test entry	Status Rejected, then a new link and Pending verification.	after reject: rejected; resubmitted with new link: true; status now pending_verification	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-133	FR-08	Reuse a used link	Link used in TC-131	1. Call Reject with the used link. 2. Open the link page.	Used token	Refused 'already been used'; page says used; entry still Verified.	action -> "This verification link has already been used."; page shows "already used": true; status still verified	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-134	FR-08	Unknown link	None	1. Call Approve with an unknown token. 2. Open its page.	48 zeros	Refused 'invalid'; page 404.	action -> "This verification link is invalid."; page status 404	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-135	FR-08	Expired link	Seeded link expired 3 days ago	1. Call Approve with the expired token. 2. Open its page.	Seeded expired token (entry-7)	Refused 'expired'; page shows expiry; entry unchanged.	action -> "This verification link has expired."; page mentions expiry: true; status pending_verification	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-136	FR-08, NFR-04	Double click on Approve	Fresh valid link	1. Send two Approve requests at the same time.	Token from TC-132	Exactly one decision recorded; the other refused.	successful decisions=1; second response "This verification link has already been used."; token decision=approved	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-137	FR-06, FR-08	Resubmit after link expired (DEF-02 retest)	DEF-02 fix deployed; entry pending with an expired link	1. Open the entry. 2. Submit again.	Entry whose link expired	Expiry notice and Submit button shown; new link issued.	page shows expiry notice and Submit button: true; resubmitted with new link: true; status pending_verification	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-138	FR-09, FR-10	Logbook view	Signed in as Sihle	1. Open /student.	Sihle's entries	Statuses shown; 'X of 200 verified hours completed' correct.	expected "12.5 of 200 verified hours"; statuses shown: Draft, Pending verification, Verified	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-139	FR-11	Monitoring dashboard	Signed in as lecturer	1. Open /coordinator.	Seed data	Dylan On track, Moe Overdue, Sihle Awaiting verification.	Dylan=On track; Moe=Overdue; Sihle=Awaiting verification; Gedion=Awaiting verification	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-140	FR-12	Automatic timestamps	TC-131 completed	1. Read the entry and link rows in the database.	TC-127 entry and link	created, verified and used times stored in ISO-8601; decision approved.	created 2026-10-22T22:54:01.841Z; verified 2026-10-22T22:54:02.000Z; link used 2026-10-22T22:54:02.000Z (approved)	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-141	FR-14	Sign out	Signed in as student	1. Click Sign out. 2. Open /student.	Student session	Cookie removed; redirected to sign-in; /student redirects.	redirect=/; cookie cleared: true; /student afterwards -> 307	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-142	NFR-03	Password storage	TC-113 account exists	1. Read the password_hash column.	TC-113 account	Salt:key hash format; plain password not stored.	stored value format: <32 hex salt>:<128 hex key>; plain password present: false	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-143	NFR-04	Database constraints	Fresh copy of schema.sql	1. Insert six kinds of invalid rows.	Hours 30, bad status, missing student, bad role, duplicate email, token without decision	Every insert rejected by a constraint.	hours = 30: rejected (CHECK constraint failed); unknown status: rejected (CHECK constraint failed); entry for missing student: rejected (FOREIGN KEY constraint failed); unknown role: rejected (CHECK constraint failed); duplicate email (case): rejected (UNIQUE constraint failed); token used without decision: rejected (CHECK constraint failed)	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt
TC-144	NFR-02	Response times	Demo data loaded	1. Request five pages 10 times each. 2. Time one entry creation.	/, /student, /student/entries/new, /coordinator, /verify/<token>	Every response under 2 seconds.	/ avg 3 ms, max 4 ms; /student avg 5 ms, max 9 ms; /student/entries/new avg 4 ms, max 5 ms; /coordinator avg 4 ms, max 6 ms; /verify/<token> avg 4 ms, max 5 ms; create entry 20 ms	Pass	Gedion Basson	2026/10/23	evidence/system_test_run_2026-10-23.txt

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TC-145	NFR-06	Mobile layout (375 px)	App running; Chrome 155 and Edge 155	1. Open sign-in, logbook, new entry, mentor page and dashboard at 375 x 812 px.	Demo data	Every screen fits the screen width; nothing cut off.	First run: logbook and dashboard tables made the page 580 px wide and cut off columns (DEF-03). Retest after fix: page width 375 px in Chrome and Edge; tables scroll inside their box.	Pass	Gedion Basson	2026/10/23	evidence/mobile_375px_chrome, evidence/mobile_375px_edge
TC-146	NFR-06, FR-14	Cross-browser sign in and sign out (Edge)	App running; Edge 155	1. Sign in as student in Edge. 2. Click Sign out. 3. Open /student.	student2@ctucareer.co.za / student123	Same behaviour as Chrome.	After sign in: /student with session cookie. After sign out: / and no cookie. /student then redirects to /.	Pass	Gedion Basson	2026/10/23	Browser check (headless Edge 155.0.4283.62)
TC-147	NFR-05	Maintainability checks	Source code on branch gedion/fix-defects	1. Run npx tsc --noEmit. 2. Run npm run lint. 3. Follow the README set-up steps.	Source code	No type or lint errors; set-up in five commands or fewer.	TypeScript: 0 errors. ESLint: 0 problems. Set-up: npm install, npm run dev (2 commands; database created automatically).	Pass	Gedion Basson	2026/10/23	README.md; terminal output
TC-148	NFR-07	Privacy review of stored data	Demo database	1. Review users and entries stored by seed.sql.	database/seed.sql	Only synthetic data and team members' own names; no staff or other students' personal data.	Seed data contains only the four team members, a synthetic coordinator and a synthetic lecturer, and invented entries; no other personal data is stored.	Pass	Gedion Basson	2026/10/23	database/seed.sql
TC-149	NFR-01	Usability: first-time student creates and submits an entry in under 3 minutes	Written campus permission and participant consent	1. Give a first-time student the task without help. 2. Time it.	Volunteer students	Task completed in under 3 minutes.	Not run: written campus permission for testing with students has not been received (Risk R01). Will be run if permission is granted before 29 October.	Blocked			Ethical Clearance Confirmation